

ORIGINAL RESEARCH

PULMONARY HYPERTENSION

Progression From Exercise-Induced to Resting Left Atrial Hypertension in HFpEF



Impact of Reduced Atriopulmonary Compliance

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ABSTRACT

BACKGROUND Approximately one-third of patients with heart failure with preserved ejection fraction (HFpEF) have normal pulmonary capillary wedge pressure (PCWP) at rest, with left atrial (LA) hypertension that is induced only during exercise (exercise-induced left atrial hypertension [EILAH]). EILAH may represent a discrete HFpEF phenotype or an earlier stage of HFpEF that progresses to resting left atrial hypertension (RELAH) over time.

OBJECTIVES The authors investigated the clinical course of HFpEF patients with EILAH who underwent repeat invasive hemodynamic evaluation.

METHODS This observational study investigated the natural history of EILAH and evaluated patients who underwent multiple invasive hemodynamic studies to assess for progression from EILAH to RELAH and then identify pathophysiological correlates of progression. Consecutively evaluated patients with HFpEF and EILAH (PCWP <15 mm Hg at rest and ≥25 mm Hg during exercise) who underwent repeat invasive hemodynamic testing were identified.

RESULTS Of 659 patients with EILAH, 84 underwent repeat invasive hemodynamic evaluations over a median follow-up of 1.81 years (Q1-Q3: 0.48-3.40 years). Of this group, 33 (39%) patients changed to RELAH (PCWP ≥15 mm Hg at rest) and 51 (61%) did not. Patients with a change to RELAH had a greater duration between their index and repeat catheterizations (2.68 years [Q1-Q3: 1.74-3.97 years] vs 0.99 years [Q1-Q3: 0.44-2.16 years]; $P = 0.001$), and they experienced greater increases in N-terminal pro-B-type natriuretic peptide (interaction $P = 0.036$), more severe worsening in left ventricular ejection fraction ($P < 0.001$), right ventricular free wall longitudinal strain ($P = 0.003$), and more worsening tricuspid regurgitation (+15% vs +2%; $P = 0.025$) compared with patients who did not transition to RELAH. Of characteristics at the index evaluation, a history of atrial fibrillation (OR: 4.57 [95% CI: 1.63-12.8]), lower LA compliance (OR: 0.47 [95% CI: 0.22-0.99]), and lower pulmonary artery compliance (OR: 0.48 [95% CI: 0.25-0.93]) were independently associated with the transition to RELAH after adjusting for follow-up duration.

CONCLUSIONS Many patients with HFpEF who first present with EILAH transition to RELAH, thus suggesting that EILAH may represent an earlier stage of HFpEF. A history of atrial fibrillation, impaired LA compliance, and reduced pulmonary artery compliance are associated with progression to RELAH. These findings call for further studies of interventions enhancing atriopulmonary compliance in early-stage HFpEF. (JACC Heart Fail. 2026;14:102947)

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS****CO** = cardiac output**eGFR** = estimated glomerular
filtration rate**EILAH** = exercise-induced left
atrial hypertension**FWLS** = free wall longitudinal
strain**GLS** = global longitudinal
strain**HF** = heart failure**HFpEF** = heart failure with
preserved ejection fraction**LA** = left atrial**LV** = left ventricular**LVEF** = left ventricular
ejection fraction**NT-proBNP** = N-terminal
pro-B-type natriuretic peptide**PA** = pulmonary artery**PAP** = pulmonary arterial
pressure**PCWP** = pulmonary capillary
wedge pressure**PVD** = pulmonary vascular
disease**RAP** = right atrial pressure**RELAH** = resting left atrial
hypertension**RV** = right ventricular**TR** = tricuspid regurgitation**V_{O2}** = oxygen uptake

Hear failure with preserved ejection fraction (HFpEF) is characterized by an inability of the heart to perfuse the body without requiring pathologic increases in cardiac filling pressure.¹⁻³ Most patients with HFpEF fulfill this definition at rest, with elevated resting pulmonary capillary wedge pressure (PCWP), but approximately one-third of patients display elevation in PCWP only during exercise.^{4,5} These patients with exercise-induced left atrial hypertension (EILAH) have an increased risk for adverse outcomes compared with patients without heart failure (HF).⁵⁻⁷ However, compared with patients with HFpEF and resting left atrial hypertension (RELAH), those with EILAH have fewer physical findings associated with congestion, lower natriuretic peptide levels, less severe cardiac remodeling and dysfunction, and a lower risk for worsening HF or death.^{4,8} These observations suggest that EILAH may represent an earlier form of HFpEF that progresses to RELAH, but this theoretical progression has not been systematically evaluated.

In longitudinal noninvasive studies, patients with HFpEF on average display worsening right-sided HF over time that is tied to the presence of atrial fibrillation,⁹ a marker of underlying atrial myopathy and a risk factor for worsening pulmonary vascular disease (PVD).¹⁰ In health, compliance

properties of the left atrium and pulmonary vessels allow for increases in volume during the cardiac cycle without untoward increases in pressure. In HFpEF, atriopulmonary compliance becomes progressively more impaired, increasing pulmonary venous pressures and pulmonary arterial pressures (PAPs) out of proportion to elevation in left ventricular (LV) diastolic pressure.

We hypothesized that a significant proportion of patients with EILAH progress to RELAH over time, and that this progression is accentuated among patients with more impaired left atrial (LA) and pulmonary artery (PA) compliance, including patients with atrial fibrillation, who are known to display the greatest reductions in atriopulmonary compliance.¹⁰ To test these hypotheses, we investigated the clinical course of patients with HFpEF and EILAH who underwent repeat invasive hemodynamic evaluations.

METHODS

STUDY GROUP AND DESIGN. Consecutive patients undergoing invasive hemodynamic testing with concurrent expiratory gas analysis to evaluate dyspnea of unknown cause at the Mayo Clinic (Rochester, Minnesota, USA), between 2006 and 2024 were identified. From this cohort, patients with HFpEF were defined by NYHA functional class II to III dyspnea with activity, left ventricular ejection fraction (LVEF) $\geq 50\%$, and PCWP ≥ 15 mm Hg at rest or ≥ 25 mm Hg during exercise. Among patients with HFpEF, EILAH was defined as PCWP < 15 mm Hg at rest and PCWP ≥ 25 mm Hg during exercise, and RELAH was defined as PCWP ≥ 15 mm Hg at rest. Patients with RELAH at the baseline evaluation, an ejection fraction $< 50\%$, primary cardiomyopathies, valvular heart disease (moderately severe or greater left-sided regurgitation, mild or greater stenosis), pericardial disease, severe coronary artery disease, congenital heart disease, and high-output HF were excluded. We then identified patients with EILAH who underwent repeat invasive hemodynamic exercise testing, who were then divided into patients whose status changed from EILAH to RELAH at the final hemodynamic evaluation and those who did not transition to RELAH. The study complies with the Declaration of Helsinki and was approved by the Institutional Review Board of the Mayo Clinic. All participants provided consent for data use by completing a research authorization form, and all authors had full access to the data and were responsible for its integrity.

ECHOCARDIOGRAPHY. Echocardiography was performed concurrently with invasive hemodynamic exercise testing. Two-dimensional, Doppler, and tissue Doppler echocardiography was performed in all patients according to the guidelines of the American Society of Echocardiography.¹¹ LV global longitudinal strain (GLS) was measured using 2-dimensional speckle tracking and determined as the average of the 2 apical views (4- and 2-chamber views). The myocardial reservoir function of the left atrium was evaluated by the LA strain peak during LA relaxation.^{10,12} Right ventricular (RV) GLS and RV free wall longitudinal strain (FWLS) were obtained by the average values of peak longitudinal systolic strain from the free wall and septal wall of the right ventricle.^{9,13} With the use of 2-dimensional echocardiographic speckle tracking, right atrial endocardial contours were traced and right atrial reservoir strain

was subsequently measured.^{14,15} All Doppler measurements were the average of ≥ 3 beats. Myocardial deformation analyses were conducted off-line (TomTec Imaging Systems), with results presented as absolute values.

CATHETERIZATION PROTOCOL. Patients underwent invasive hemodynamic exercise testing with simultaneous expiratory gas analysis to volitional exhaustion by supine cycle ergometry, by using methods we have previously described.^{16,17} Right atrial pressure (RAP), PAP, and PCWP were measured at end-expiration (average of ≥ 3 beats) by using a 2.0-F high-fidelity micromanometer-tipped catheter advanced through the lumen of a 7.0-F fluid-filled catheter at rest. During exercise, PAP and PCWP were measured using a micromanometer, and RAP was measured from the side arm of the 9.0-F sheath at the junction of the superior vena cava and the right atrium. The position of PCWP was confirmed by standard criteria, such as appearance on fluoroscopy, characteristic pressure waveforms, and evidence of oxyhemoglobin saturation $\geq 94\%$. A 4.0-F to 6.0-F radial arterial cannula was used to measure arterial blood pressure and obtain arterial blood gas samples throughout the test. Pressure tracings from the entire study were digitized (240 Hz) and stored for off-line analysis.

After baseline hemodynamic assessment, study subjects underwent supine cycle exercise testing, with the first stage at 20 W for 3-5 minutes, followed by increasing workload in 20-W increments until volitional exhaustion (2-minute stages, with hemodynamics assessed in the last minute of each stage). As with rest, pressures were measured at end-expiration by 1 observer as an average of 3 cardiac cycles (or 5 cardiac cycles in the case of atrial fibrillation). Arterial and mixed venous (PA) oxygen samples were taken at baseline, at 20 W, and during peak exercise. Oxygen uptake ($\dot{V}O_2$) was calculated by expiratory gas analysis (MedGraphics). The arterial-venous oxygen content difference ($Ca-V_{O_2}$) was directly measured as the difference between systemic arterial and PA oxygen contents ($= \text{saturation} \times \text{hemoglobin} \times 1.34$). Cardiac output (CO) was determined by the direct Fick method ($CO = \dot{V}O_2 / Ca-V_{O_2}$). Pulmonary vascular resistance was calculated by: $(\text{mean PAP} - \text{PCWP}) / CO$. Pulmonary artery compliance was calculated by stroke volume/PA pulse pressure. LA compliance was calculated as the quotient of LA reservoir strain and the difference between the V-wave peak and the x-descent nadir.¹⁰ Systemic vascular resistance was calculated by: $(\text{mean arterial pressure} - \text{RAP}) / CO$.

STATISTICAL ANALYSIS. Continuous variables are presented as mean $\pm$ SD when normally distributed and as median (Q1-Q3) when non-normally distributed. Parameters were compared between groups that changed from EILAH to RELAH and those who did not transition to RELAH by using unpaired Student's *t*-tests or Mann-Whitney *U*-tests for continuous variables and by chi-squared tests or the Fisher exact test for dichotomous variables, when appropriate. Within-group differences at individual time points between the baseline and the final examination were compared using a paired Student's *t*-test. Within-group differences were assessed by 1-way repeated measures analysis of variance to account for the rest, 20-W exercise, and peak exercise. To account for unequal follow-up duration and missing data, longitudinal changes between baseline and follow-up examinations were evaluated using linear mixed-effects models. The participant was included as a random effect to account for within-subject correlation. Fixed effects included group (transition to RELAH vs no transition), visit (baseline vs follow-up), time between assessments, and the group-by-visit interaction. Absolute changes in characteristics, laboratories, echocardiography and hemodynamics were calculated by subtracting the final visit values from the baseline visit values to define the dependent variables for the analysis. Multivariable logistic regression analysis was performed to assess determinants of transition from EILAH to RELAH. Candidate predictors were selected a priori on the basis of clinical plausibility and pathophysiologic relevance, with univariable analyses used to inform but not dictate model inclusion. To minimize overfitting given the limited number of events, a maximum of 3 noncolinear covariates were included in each model. A 2-sided value of $P < 0.05$ was considered statistically significant. The predictive ability for transition from EILAH to RELAH was evaluated by area under the curve. All data were analyzed using Stata MP64 software version 16 (StataCorp).

RESULTS

PATIENT CHARACTERISTICS AT BASELINE EXAMINATION.

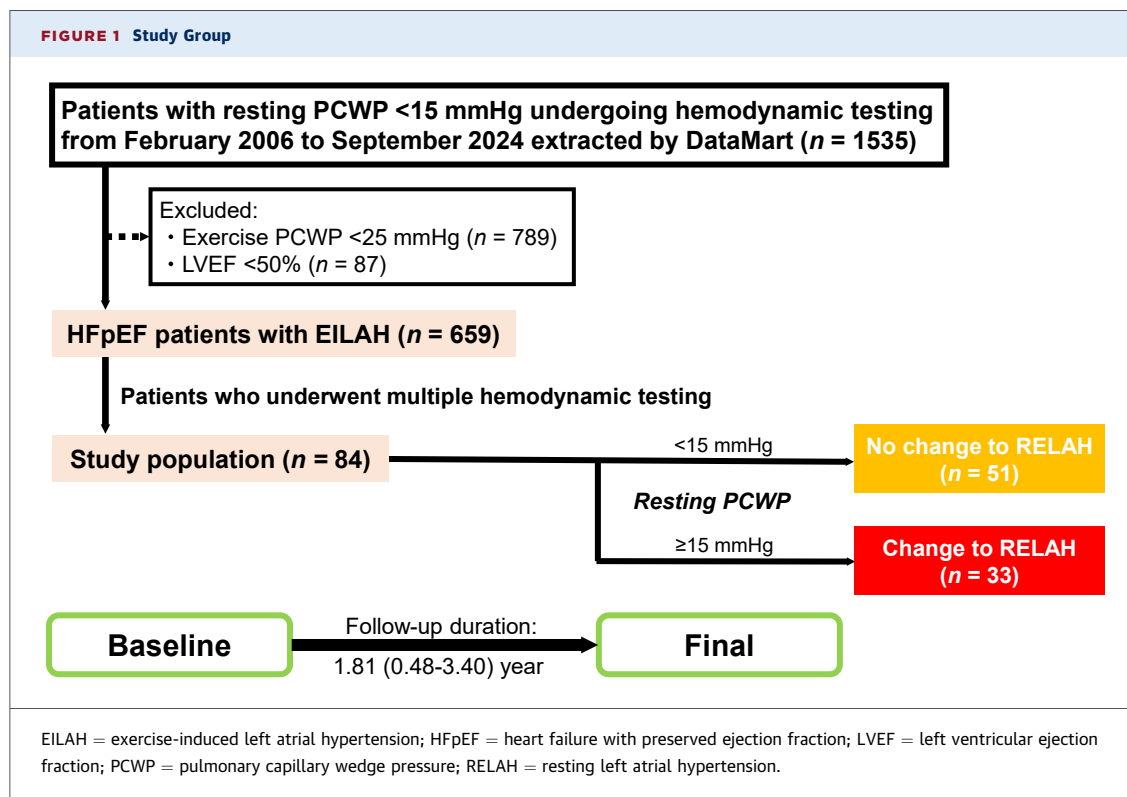
Of 1,535 consecutively evaluated patients undergoing hemodynamic exercise testing with resting PCWP < 15 mm Hg, 789 (51%) had exercise PCWP < 25 mm Hg, and 87 (6%) had an ejection fraction $< 50\%$, leaving 659 (43%) patients with EILAH. Of this group, 84 (13%) underwent repeat invasive exercise testing a median of 1.81 years (Q1-Q3: 0.48-3.40 years) following the index study. The most

TABLE 1 Patients' Characteristics at Baseline Examination

	Total (N = 84)	No Change to RELAH (n = 51)	Change to RELAH (n = 33)	P Value
Age, y	66 ± 10	65 ± 10	67 ± 11	0.43
Women	55 (65)	30 (59)	25 (76)	0.11
Body weight, kg	93.0 ± 22.2	95.6 ± 22.3	89.0 ± 21.7	0.18
BMI, kg/m ²	33.1 ± 6.8	33.7 ± 6.9	32.3 ± 6.5	0.37
H ₂ FPEF score	4 (3-6)	4 (3-6)	5 (3-6)	0.48
Indications for repeated testing, %				
Symptom/clinical follow-up/study visit	40/13/46	33/12/55	52/15/33	0.15
Comorbidities				
Obesity (BMI ≥30 kg/m ²)	57 (68)	36 (71)	21 (64)	0.51
Hypertension	60 (71)	36 (71)	24 (73)	0.83
Diabetes	27 (32)	19 (37)	8 (24)	0.21
Any atrial fibrillation	29 (35)	11 (22)	18 (55)	0.002
Paroxysmal/persistent AF	22/7 (76/24)	10/1 (91/9)	12/6 (67/33)	0.14
Stage ≥3 CKD	28 (33)	14 (27)	14 (42)	0.16
Coronary artery disease	48 (57)	31 (61)	17 (52)	0.40
COPD	7 (8)	3 (6)	4 (12)	0.31
Sleep apnea syndrome	41 (49)	24 (47)	17 (52)	0.69
CIED	13 (15)	5 (10)	8 (24)	0.074
Pulmonary vein isolation	11 (13)	5 (10)	6 (18)	0.27
Medications				
Diuretic agent	38 (45)	21 (41)	17 (52)	0.35
RAAS inhibitor	37 (44)	22 (43)	15 (45)	0.83
Beta-blocker	41 (49)	25 (49)	16 (48)	0.96
Mineralocorticoid antagonist	25 (30)	18 (35)	7 (21)	0.17
SGLT2 inhibitor	5 (6)	4 (8)	1 (3)	0.36
Laboratory values				
Hemoglobin, g/dL	12.8 ± 1.6	13.0 ± 1.6	12.5 ± 1.7	0.21
eGFR, mL/min/1.73 m ²	67.2 (55.7-77.9)	71.8 (59.5-81.7)	60.9 (49.2-71.9)	0.023
NT-proBNP, pg/mL	173 (77-387)	119 (57-297)	289 (99-781)	0.010
Echocardiography				
LVEF, %	60 ± 7	60 ± 7	61 ± 7	0.50
LV diastolic dimension, mm	48.9 ± 5.3	49.5 ± 5.4	48.0 ± 5.0	0.21
LV systolic dimension, mm	31.3 ± 4.8	31.7 ± 4.7	30.7 ± 4.9	0.36
LA volume index, mL/m ²	32.5 (24.5-44.0)	28.8 (23.0-40.5)	37.0 (29.5-54.0)	0.009
LV mass index, g/m ²	89.9 ± 29.5	87.4 ± 24.8	94.1 ± 36.0	0.33
E-wave velocity, m/s	0.7 (0.6-0.9)	0.7 (0.6-0.9)	0.7 (0.6-1.0)	0.077
E/A ratio	0.88 (0.71-1.21)	0.87 (0.71-1.19)	0.98 (0.79-1.42)	0.53
Septal e' velocity, m/s	0.07 ± 0.02	0.07 ± 0.02	0.07 ± 0.02	0.64
Lateral e' velocity, m/s	0.09 ± 0.02	0.09 ± 0.02	0.09 ± 0.02	0.77
Average E/e'	10.0 (8.0-11.7)	10.0 (7.8-11.4)	10.0 (8.3-13.3)	0.29
TR velocity, m/s	2.59 (2.40-2.80)	2.52 (2.40-2.70)	2.69 (2.37-3.07)	0.11
LV GLS, %	15.4 ± 4.3	15.9 ± 4.2	14.6 ± 4.4	0.20
LA reservoir strain, %	24.6 ± 9.6	26.7 ± 8.5	21.1 ± 10.3	0.013
LA booster strain, %	12.4 ± 6.2	12.8 ± 5.9	11.6 ± 7.0	0.50
RV GLS, %	16.6 ± 6.1	16.8 ± 5.8	16.2 ± 6.6	0.64
RV FWLS, %	18.6 ± 7.3	18.3 ± 7.0	19.1 ± 7.8	0.63
RA reservoir strain, %	22.4 ± 10.6	22.4 ± 9.2	22.5 ± 12.6	0.96
RA booster strain, %	11.0 ± 6.1	10.8 ± 4.7	11.2 ± 8.6	0.87
Any MR	14 (17)	7 (14)	7 (21)	0.37
Any TR	38 (45)	20 (39)	18 (55)	0.17
Mild atrial MR	12 (14)	6 (12)	6 (18)	0.41
Mild TR	31 (37)	18 (35)	13 (39)	0.70
Moderate or severe MR	2 (2)	1 (2)	1 (3)	0.75
Moderate or severe TR	7 (8)	2 (4)	5 (15)	0.069

Values are mean ± SD, n (%), median (Q1-Q3), or n/N (%/%), unless otherwise indicated.

AF = atrial fibrillation; BMI = body mass index; CIED = cardiovascular implantable electronic device; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; eGFR = estimated glomerular filtration rate; FWLS = free wall longitudinal strain; GLS = global longitudinal strain; H₂FPEF (score) = age, hypertension, obesity, atrial fibrillation status, and resting Doppler echocardiographic estimated pulmonary artery systolic pressure or left atrial pressure; LA = left atrial; LV = left ventricular; LVEF = left ventricular ejection fraction; MR = mitral regurgitation; NT-proBNP = N-terminal pro-B-type natriuretic peptide; RA = right atrial; RAAS = renin-angiotensin-aldosterone system; RELAH = resting left atrial hypertension; RV = right ventricular; SGLT2 = sodium-glucose cotransporter 2; TR = tricuspid regurgitation.



common indication for repeat catheterization was participation in a research study that included cardiac catheterization as part of its protocol (46%), followed by worsening symptoms (40%) and other clinical follow-up not further specified (14%) (Table 1). Compared with patients with EILAH who did not undergo repeat invasive testing, those who underwent follow-up evaluation were more likely to have a history of atrial fibrillation and diabetes, to be prescribed diuretic agents, and to have slightly poorer kidney function (Supplemental Table 1). In contrast, key markers of LA structure and diastolic function, including LA volume index, E/e', tricuspid regurgitation (TR) velocity, and N-terminal pro-B-type natriuretic peptide (NT-proBNP), were similar between groups (Supplemental Table 1).

Of the 84 patients who underwent repeat testing, 33 (39%) transitioned to RELAH at repeat evaluation, whereas 51 (61%) did not (Figure 1). The interval from the first to the final hemodynamic evaluation was significantly longer in patients with a change to RELAH (2.68 years [Q1-Q3: 1.74-3.97 years] vs 0.99 years [Q1-Q3: 0.44-2.16 years]; $P = 0.001$). There was no significant difference between the 2 groups in the indication for repeat catheterization as a result of worsening symptoms (52% vs 33%; $P = 0.15$). Patients with a subsequent change to RELAH had a higher

prevalence of atrial fibrillation at baseline, but there were no differences in age, sex, body weight, body mass index, prevalence of obesity, hypertension, diabetes, coronary artery disease, or medication use. Patients with a change to RELAH had a lower estimated glomerular filtration rate (eGFR) and higher NT-proBNP levels at baseline (Table 1). At the baseline echocardiogram, patients with a subsequent change to RELAH displayed higher LA volume index, lower LA reservoir strain, and higher prevalence of TR, but there were no differences in LVEF, LV mass index, average E/e', TR velocity, and prevalence of MR when compared with patients with no change to RELAH (Table 1).

On baseline hemodynamic evaluation, patients displayed typical findings of HFpEF with EILAH, including normal resting biventricular filling pressures and normal resting CO (Table 2). Pulmonary hypertension at rest (mean PAP >20 mm Hg) was present at the index evaluation in 27 (53%) of patients who did not progress to RELAH and in 23 (70%) of patients who did progress to RELAH ($P = 0.13$ between groups). Compared with patients who did not transition to RELAH at follow-up, those with a change to RELAH displayed a higher resting heart rate and systolic PAP, and lower PA and LA compliance (Figure 2), but there were no differences in

TABLE 2 Invasive Hemodynamic Assessment at Baseline Examination

	Total (N = 84)	No Change to RELAH (n = 51)	Change to RELAH (n = 33)	P Value
Rest				
Heart rate, beats/min	71 ± 12	68 ± 11	75 ± 13	0.006
Systolic BP, mm Hg	141 ± 22	140 ± 19	143 ± 26	0.58
Central pressures				
RAP, mm Hg	7 ± 3	7 ± 3	7 ± 3	0.88
PCWP, mm Hg	11 ± 2	11 ± 2	12 ± 2	0.69
PCWP v-wave, mm Hg	17 ± 5	16 ± 4	18 ± 6	0.10
PCWP v-wave height, mm Hg	6 ± 4	5 ± 4	7 ± 4	0.064
LA compliance, %/mm Hg	4.34 (2.40-8.00)	5.12 (3.20-8.07)	2.60 (1.67-7.40)	0.004
Systolic PAP, mm Hg	34 ± 7	32 ± 6	35 ± 7	0.049
Mean PAP, mm Hg	21 ± 4	20 ± 4	22 ± 5	0.11
Mean PAP >20 mm Hg	50 (60)	27 (53)	23 (70)	0.13
Vascular function				
PVR, WU	1.84 (1.20-2.45)	1.79 (0.98-2.26)	1.89 (1.31-2.87)	0.12
PVR >2.0 WU	35 (42)	20 (39)	15 (45)	0.57
PA compliance, mL/mm Hg	4.1 ± 1.5	4.4 ± 1.5	3.5 ± 1.3	0.006
Oxygen transport				
Vo ₂ , mL/min/kg	229 ± 48	231 ± 49	227 ± 48	0.70
Ca-Vo ₂ , mL/dL	4.6 ± 1.1	4.5 ± 1.2	4.7 ± 1.0	0.66
CO, L/min	5.16 ± 1.34	5.25 ± 1.43	5.02 ± 1.19	0.43
Stroke volume, mL	74.1 ± 20.0	78.4 ± 20.9	67.5 ± 17.0	0.014
Exercise				
Heart rate, beats/min	110 ± 21	109 ± 21	112 ± 22	0.49
Systolic BP, mm Hg	188 ± 33	189 ± 33	187 ± 33	0.85
Peak workload, W	60 (40-80)	60 (60-80)	40 (20-80)	0.006
RER	1.04 (0.97-1.10)	1.06 (0.96-1.12)	1.03 (0.97-1.09)	0.59
Central pressures				
RAP, mm Hg	16 ± 5	15 ± 5	16 ± 5	0.37
PCWP, mm Hg	31 ± 6	30 ± 5	33 ± 8	0.085
Systolic PAP, mm Hg	61 ± 13	58 ± 12	65 ± 15	0.016
Mean PAP, mm Hg	40 ± 9	39 ± 8	43 ± 9	0.031
PCWP/CO slope	4.4 (2.9-6.9)	3.7 (2.8-5.8)	5.9 (4.2-10.2)	0.004
Mean PAP/CO slope	4.5 (2.6-7.4)	4.1 (2.5-5.6)	6.0 (3.8-12.2)	0.003
Vascular function				
PVR, WU	0.76 (0.32-1.44)	0.67 (0.13-1.18)	0.90 (0.47-2.00)	0.053
PA compliance, mL/mm Hg	3.1 ± 1.4	3.5 ± 1.5	2.6 ± 1.1	<0.001
Oxygen transport				
Vo ₂ , mL/min/kg	1,031 ± 335	1,089 ± 324	931 ± 335	0.043
Ca-Vo ₂ , mL/dL	10.8 ± 2.2	10.8 ± 2.4	10.8 ± 2.0	1.00
CO, L/min	9.70 ± 2.74	10.27 ± 2.57	8.74 ± 2.80	0.016
Stroke volume, mL	90.3 ± 29.6	97.7 ± 31.9	77.8 ± 20.0	0.003

Values are mean ± SD, median (Q1-Q3), or n (%), unless otherwise indicated.

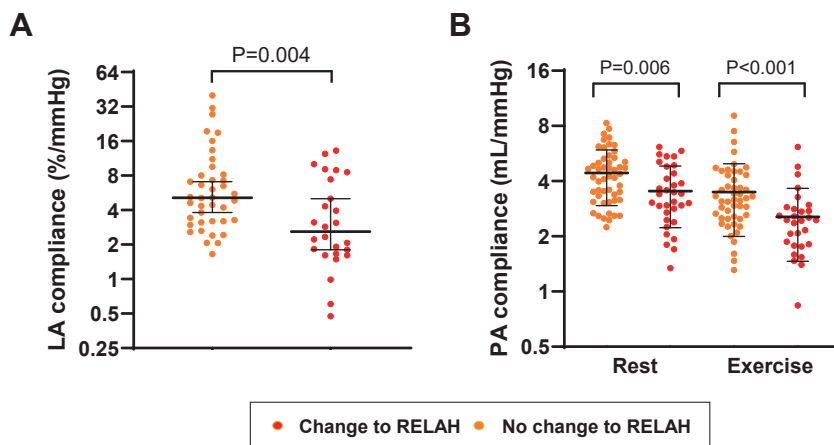
BP = blood pressure; Ca-Vo₂ = arterial-venous oxygen content difference; CO = cardiac output; PA = pulmonary artery; PAP = pulmonary artery pressure; PCWP = pulmonary capillary wedge pressure; PVR = pulmonary vascular resistance; RAP = right atrial pressure; RER = respiratory exchange ratio; SVR = systemic vascular resistance; Vo₂ = oxygen consumption; other abbreviations as in [Table 1](#).

resting RAP, PCWP, or CO between the groups ([Table 2](#)). At peak exercise during the baseline hemodynamic evaluation, patients with a transition to RELAH displayed higher systolic PAP, mean PAP, higher PCWP/CO slope and mean PAP/CO slope, and lower peak workload, peak Vo₂, PA compliance, CO, and stroke volume ([Table 2, Figure 2B](#)). The invasive hemodynamics at final examination are shown in [Supplemental Table 2](#).

In a sensitivity analysis limited to patients who underwent repeat catheterization as part of a research protocol (n = 39), the rate of transition to RELAH was similar (28%), and the direction of associations for key predictors was consistent with the main analysis ([Supplemental Tables 3 and 4](#)).

LONGITUDINAL CHANGES IN HEMODYNAMICS AND CLINICAL MEASURES. At the final examination, there were significant decreases in both eGFR and

FIGURE 2 Differences in LA Compliance and PA Compliance at Baseline



LA compliance at rest (A) and PA compliance at rest and during exercise (B) in patients with a change to RELAH or who did not transition to RELAH. Error bars reflect SD. LA = left atrial; PA = pulmonary artery; other abbreviation as in Figure 1.

body weight observed among all patients (both $P < 0.01$ for within-group changes), but no differences in these changes between those patients who transitioned to RELAH and those who did not (Table 3, Figure 3). Patients with a change to RELAH had more frequently been started on beta-blockers (27% vs 4%; $P = 0.002$), but there were no differences between the groups in other medication use, including diuretic agents (Supplemental Table 5). Compared with patients who did not change to RELAH, those with a transition to RELAH had a greater increase in NT-proBNP level (interaction $P = 0.036$), had a greater decrease in LVEF ($P < 0.001$; even as LVEF remained normal), were more likely to experience worsening TR severity (15% vs 2%; $P = 0.025$), and had greater decreases in RV GLS ($P = 0.010$) and RV FWLS ($P = 0.003$) (Table 3, Figure 3).

On final invasive hemodynamic evaluation at rest, patients with a change to RELAH displayed a lower resting heart rate (interaction $P < 0.001$), increased RAP ($P < 0.001$), and increased mean PAP ($P < 0.001$) compared with patients with no change (Table 3, Figure 4). At peak exercise, there were no differences in PCWP, RAP, mean PAP, or CO (Table 3, Figure 4).

CORRELATES OF THE CHANGE FROM EILAH TO RELAH IN BASELINE PARAMETERS. Transition from EILAH to RELAH was most strongly predicted by lower resting LA compliance and lower exercise PA compliance (area under the curve C-statistics: 0.71 [95% CI: 0.57-0.85] and 0.72 [95% CI: 0.60-0.84],

respectively) (Supplemental Table 6). In univariable logistic regression, a history of atrial fibrillation, a longer duration of follow-up, higher resting heart rate, NT-proBNP, and LA volume index, and lower LA reservoir strain, LA compliance, PA compliance, peak workload, and peak Vo_2 were associated with a change from EILAH to RELAH (Table 4). In multivariable logistic regression of baseline parameters, a history of atrial fibrillation (OR: 4.57 [95% CI: 1.63-12.8]; $P = 0.004$), lower resting LA compliance (OR: 0.47 [95% CI: 0.22-0.99]; $P = 0.049$), and lower resting PA compliance (OR: 0.48 [95% CI: 0.25-0.93]; $P = 0.030$) were independently associated with a change from EILAH to RELAH even after adjusting the duration from baseline to final examination (Table 4).

DISCUSSION

Approximately one-third of patients with HFpEF have an elevation in PCWP only during exercise (EILAH), and it has remained unclear whether such patients represent a distinct form of HF, or potentially, an earlier stage that may progress to HFpEF that is apparent at rest (RELAH). Using serial invasive hemodynamic evaluations, we show here for the first time that many patients who first presented with EILAH transitioned to RELAH at repeat assessment. Patients with a change to RELAH had lower eGFR, higher NT-proBNP, more TR, higher LA volume index, and lower LA reservoir strain, but the strongest predictors of transition to RELAH were impairments

TABLE 3 Longitudinal Changes From Baseline to Final Examination

	No Change to RELAH	P Value	Change to RELAH	P Value	Interaction P Value
Characteristics					
Change in body weight, kg	−2.8 (−5.0 to −0.6)	0.013	−5.7 (−8.5 to −3.0)	<0.001	0.10
Diuretic initiation or dose increase	16 (31)	—	12 (36)	—	—
Mineralocorticoid antagonist initiation	13 (25)	—	9 (27)	—	—
Beta-blocker initiation	2 (4)	—	9 (27)	—	—
SGLT2 inhibitor initiation	13 (25)	—	6 (18)	—	—
New onset of AF	5 (10)	—	2 (6)	—	—
New intervention for coronary artery	0	—	0	—	—
Laboratory values					
Change in hemoglobin, g/dL	0.1 (−0.2 to +0.5)	0.51	−0.1 (−0.5 to +0.3)	0.65	0.45
Change in eGFR, mL/min/1.73 m ²	−6.2 (−11.0 to −1.4)	0.01	−11.1 (−17.1 to −5.1)	<0.001	0.21
Change in NT-proBNP, pg/mL	121 (−1,879 to +2,121)	0.90	3,575 (1,053-6,097)	0.006	0.036
Echocardiography					
Change in LVEF, %	1.0 (−1.2 to +3.2)	0.38	−5.3 (−8.1 to −2.6)	<0.001	<0.001
Change in LA volume index, mL/m ²	−1.8 (−5.8 to +2.2)	0.37	3.2 (−2.5 to +8.9)	0.27	0.16
Change in LV mass index, g/m ²	−4.2 (−12.2 to +3.7)	0.29	−7.6 (−17.5 to +2.3)	0.13	0.60
Change in TR velocity, m/s	−0.03 (−0.17 to +0.11)	0.68	0.03 (−0.15 to +0.21)	0.75	0.61
Worsening TR severity	1 (2)	—	5 (15)	—	—
Change in LV GLS, %	1.0 (−0.3 to +2.4)	0.15	−0.9 (−2.6 to +0.8)	0.28	0.08
Change in LA reservoir strain, %	−2.30 (−5.1 to +0.5)	0.11	−2.4 (−6.1 to +1.3)	0.19	0.96
Change in RV GLS, %	2.1 (−0.4 to +4.5)	0.10	−3.2 (−6.4 to −0.10)	0.043	0.010
Change in RV FWLS, %	2.6 (−0.7 to +5.9)	0.13	−5.7 (−9.9 to −1.5)	0.009	0.003
Change in RA reservoir strain, %	0.9 (−4.3 to +6.1)	0.73	−7.2 (−12.9 to −1.4)	0.015	0.04
Hemodynamics at rest					
Change in systolic BP, mm Hg	0.8 (−5.6 to +7.2)	0.80	−7.2 (−15.2 to +0.7)	0.07	0.12
Change in heart rate, beats/min	4.2 (−0.8 to +7.6)	0.016	−5.0 (−9.2 to −0.8)	0.022	<0.001
Change in PCWP, mm Hg	−1.2 (−2.2 to −0.1)	0.028	6.7 (5.4-8.0)	<0.001	<0.001
Change in RAP, mm Hg	−1.5 (−2.6 to −0.4)	0.052	2.9 (1.5-4.3)	<0.001	<0.001
Change in mean PAP, mm Hg	−2.4 (−3.9 to −0.9)	0.002	4.2 (2.3-6.0)	<0.001	<0.001
Change in PVR, WU	−0.25 (−0.59 to +0.06)	0.11	−0.34 (−0.73 to +0.03)	0.07	0.69
Change in PA compliance, mL/mm Hg	0.09 (−0.43 to +0.62)	0.73	−0.32 (−0.97 to +0.34)	0.34	0.34
Change in CO, L/min	−0.15 (−0.54 to +0.24)	0.44	−0.53 (−1.02 to −0.05)	0.03	0.22
Hemodynamics during peak exercise					
Change in systolic BP, mm Hg	−8.9 (−21.3 to +3.4)	0.15	−6.5 (−23.6 to +10.5)	0.45	0.82
Change in heart rate, beats/min	2.0 (−3.8 to +7.7)	0.50	−2.7 (−10.8 to +5.4)	0.51	0.35
Change in PCWP, mm Hg	−7.6 (−10.0 to −5.3)	<0.001	−4.5 (−7.8 to −1.3)	0.007	0.13
Change in RAP, mm Hg	−1.5 (−3.7 to +0.6)	0.15	1.9 (−1.1 to +4.9)	0.21	0.07
Change in mean PAP, mm Hg	−4.4 (−7.1 to −1.8)	0.002	−0.4 (−4.2 to +3.3)	0.82	0.09
Change in PVR, WU	0.33 (0.07-0.60)	0.013	0.65 (0.28-1.03)	0.001	0.18
Change in PA compliance, mL/mm Hg	0.62 (0.17-1.08)	0.008	−1.07 (−1.90 to −0.25)	0.011	0.36
Change in CO, L/min	0.17 (−0.69 to +1.03)	0.70	−0.15 (−1.39 to +1.09)	0.81	0.68

Values are n (%) or mean (95% CI) estimated using linear mixed-effects models, unless otherwise indicated. Models included participant as a random effect and group, visit, time between assessments, and group-by-visit interaction as fixed effects.

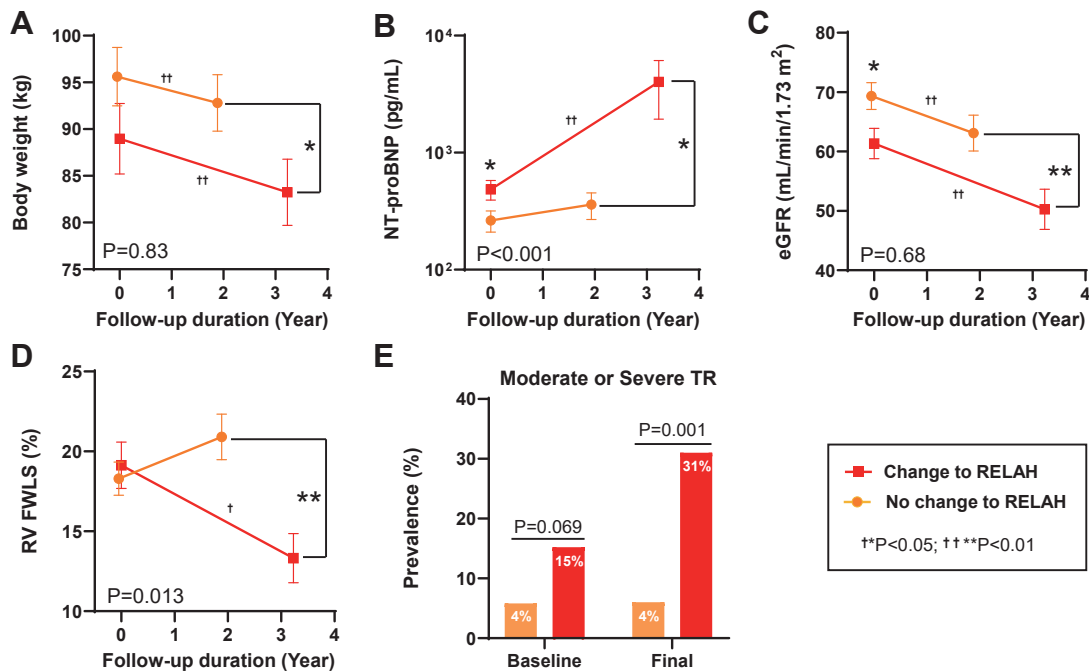
Abbreviations as in [Tables 1 and 2](#).

in LA and PA compliance at baseline, along with prevalent atrial fibrillation, which is intimately related to atrial myopathy and PVD in HFpEF. Patients who transitioned to RELAH had a greater increase in NT-proBNP level and were more likely to experience worsening TR severity and a greater decrease in biventricular systolic function. The present findings support the idea that EILAH may represent an earlier stage of HFpEF and call for

testing whether interventions that enhance atrio-pulmonary compliance could mitigate HFpEF progression ([Central Illustration](#)).

CHARACTERISTICS OF PATIENTS WITH EILAH WHO PROGRESS TO RELAH. NT-proBNP levels, prevalence of atrial fibrillation, and prevalence of chronic kidney disease were relatively low in the patients with EILAH at the index evaluation, findings consistent with those of previous cross-sectional studies.^{5,8}

FIGURE 3 Longitudinal Change of Patients' Characteristics



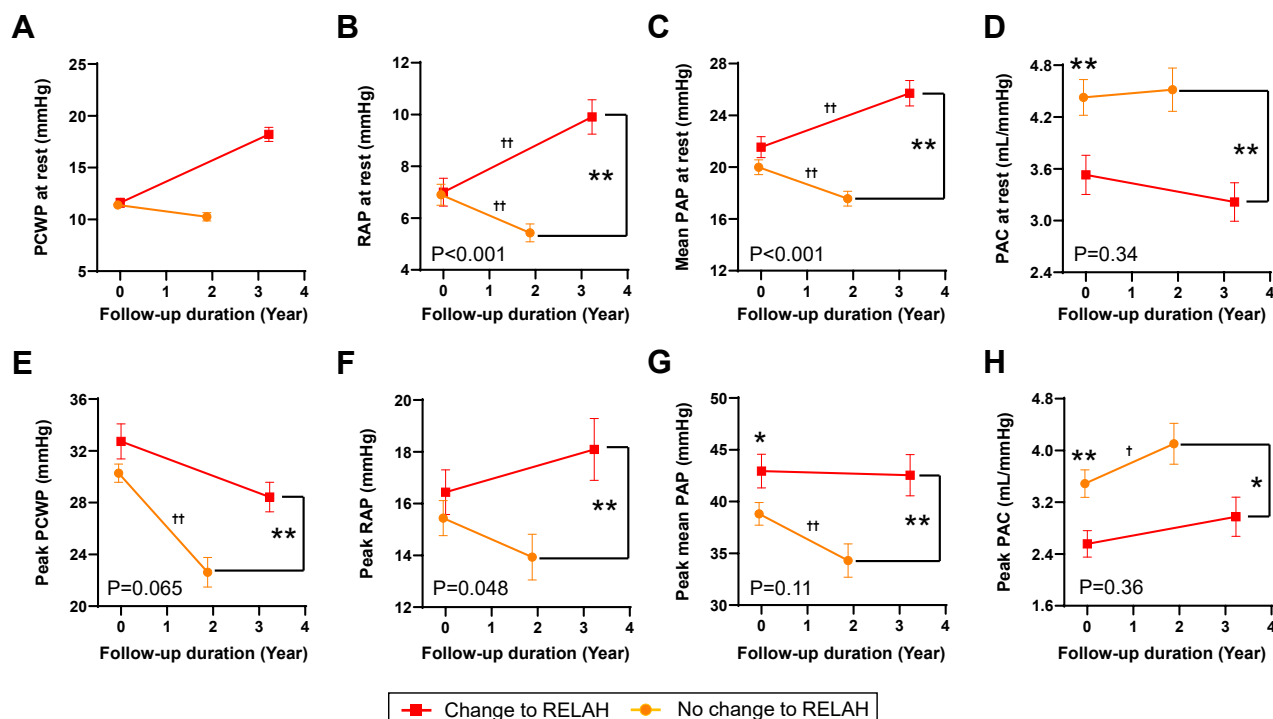
Body weight (A), NT-proBNP (B), eGFR (C), RV FWLS (D), and prevalence of TR (E) in patients with a change to RELAH (red) or who did not transition to RELAH (orange). Error bars reflect SE. *Mean single time point between group comparisons: * $P < 0.05$, ** $P < 0.01$. †Means paired Student's t -test for within-group comparisons from baseline to final examinations: † $P < 0.05$, †† $P < 0.01$. eGFR = estimated glomerular filtration rate; FWLS = free wall longitudinal strain; NT-proBNP = N terminal pro brain natriuretic peptide; RV = right ventricular; TR = tricuspid regurgitation; other abbreviation as in Figure 1.

Patients displayed typical findings of HFpEF with EILAH,^{5,8} including normal resting biventricular filling pressures and CO. Most patients displayed pulmonary hypertension at baseline, which, in the absence of exercise testing, would have been considered as precapillary pulmonary hypertension rather than HFpEF because the resting PCWP was normal and potentially treated in at least some individuals with pulmonary vasodilator agents rather than treatments for HFpEF.

The most significant finding of this study is that a substantial proportion of patients initially presenting with EILAH transitioned to RELAH over time, thus suggesting that EILAH represents an earlier stage of HFpEF. Interestingly, the patients with a change to RELAH displayed markers of greater atrial myopathy, myocardial stress, and cardiorenal abnormalities, identifying a subgroup within EILAH with a higher risk of disease progression. These patients had greater LA enlargement, reduced LA reservoir strain, and lower LA compliance at baseline, as well as higher prevalence of functional TR, an important

clue to the presence of underlying HFpEF.^{18,19} In a smaller study of 45 patients with EILAH, 5 were found to have RELAH at subsequent hemodynamic evaluations (11%), although it was not reported how many of the 45 patients underwent repeat catheterization.⁶ The present findings align with previous studies indicating that atrial remodeling plays a pivotal role in the pathophysiology of HFpEF and may contribute to the eventual development of PVD.¹⁰ Longitudinal changes further highlight the progressive natural history of this condition. Patients with a transition to RELAH exhibited a greater increase in NT-proBNP levels over time, alongside worsening TR severity and significant declines in RV GLS and RV FWLS. These findings suggest that worsening RV function and increased filling pressures may be key contributors to or consequences of transition from EILAH to RELAH. Although our main analysis focused on parameters of baseline examination, it is noteworthy that patients who progressed to RELAH displayed higher exercise pulmonary vascular resistance at follow-up (median 1.65 WU)

FIGURE 4 Longitudinal Change of Invasive Hemodynamics



At resting hemodynamics, PCWP (A), RAP (B), mean PA pressure (C), and PA compliance (D); at peak exercise, PCWP (E), RAP (F), mean PA pressure (G), and PA compliance (H) in patients with a change to RELAH (red) or who did not transition to RELAH (orange). Error bars reflect SE. *Mean single time point between group comparisons: *P < 0.05, **P < 0.01. †Means paired Student's t-test for within-group comparisons from baseline to final examinations: †P < 0.05, ††P < 0.01. RAP = right atrial pressure; other abbreviations as in Figures 1 and 2.

(Supplemental Table 2), a value approaching the 1.74-WU threshold that identified responders in the REDUCE-LAP II (A Study to Evaluate the Corvia Medical, Inc IASD System II to Reduce Elevated Left Atrial Pressure in Patients With Heart Failure) trial.²⁰ This observation further supports the notion that pulmonary vascular involvement becomes manifest as the disease advances from EILAH to RELAH.

IDENTIFYING RISK AND PREVENTING TRANSITION TO RELAH. Compared with patients with noncardiac dyspnea, patients with HFpEF and EILAH have an increased risk for HF hospitalization or death, but patients with elevated resting PAWP (RELAH) have an even higher risk than those with EILAH.⁵ Therefore, although both EILAH and RELAH are part of the HFpEF spectrum, it may be important to identify patients with EILAH at higher risk for transition to RELAH to initiate treatment. The present results relating worsening atrial and pulmonary vascular compliance, as well as markers of right-sided HF, as

predictors of this transition are consistent with previous noninvasive longitudinal studies from Obokata et al⁹ and Sorimachi et al²¹ that showed deterioration in RV function, increasing TR, and greater impairment in LA reservoir strain over time in HFpEF.

Arterial, pulmonary arterial, and atrial compliances reflect the abilities of the systemic and pulmonary circulations and left atrium (respectively) to accept the stroke volume and venous return during the cardiac cycle without untoward increases in pressure. In the present study, PA compliance was lower in those patients who later developed RELAH, thereby suggesting that loss of this buffering capacity of the lung vessels could contribute to progression to resting LA hypertension. The sodium-glucose cotransporter 2 (SGLT2) inhibitor dapagliflozin has been shown to improve PA compliance at rest and during exercise in HFpEF,²² along with improvements in total arterial compliance.²³ SGLT2 inhibitors may also favorably affect atrial myopathy because secondary analyses from the DECLARE-TIMI 58

TABLE 4 Predictors of the Transition to RELAH From Baseline Examination

	Univariable Analysis		Multivariable Analysis			
			Model 1		Model 2	
	OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Follow-up duration, y	1.71 (1.07-2.73)	0.025	1.49 (0.88-2.51)	0.14	1.21 (0.72-2.05)	0.48
Age, y	1.19 (0.77-1.85)	0.43	Not selected	—	Not selected	—
Female	2.18 (0.83-5.78)	0.11	Not selected	—	Not selected	—
Body weight, kg	0.73 (0.46-1.16)	0.18	Not selected	—	Not selected	—
Systolic BP, mm Hg	1.13 (0.73-1.75)	0.57	Not selected	—	Not selected	—
History of obesity	0.73 (0.29-1.85)	0.51	Not selected	—	Not selected	—
Hemoglobin, mg/dL	0.74 (0.46-1.19)	0.21	Not selected	—	Not selected	—
Rest mean PAP, mm Hg	1.59 (0.89-2.82)	0.12	Not selected	—	Not selected	—
Log rest PVR, WU	1.68 (0.93-3.00)	0.088	Not selected	—	Not selected	—
Heart rate, beats/min	1.91 (1.17-3.11)	0.009	Not selected	—	Not selected	—
History of AF	4.36 (1.68-11.4)	0.003	4.57 (1.63-12.8)	0.004	Not selected	—
Log NT-proBNP, pg/mL	1.83 (1.12-2.98)	0.016	Not selected	—	Not selected	—
LA volume index, mL/m ²	1.96 (1.28-3.41)	0.017	Not selected	—	Not selected	—
LA reservoir strain, %	0.54 (0.33-0.91)	0.020	Not selected	—	Not selected	—
Log LA compliance, %/mm Hg	0.55 (0.18-0.75)	0.006	Not selected	—	0.47 (0.22-0.99)	0.049
Log rest PA compliance, mL/mm Hg	0.46 (0.26-0.82)	0.009	0.48 (0.25-0.93)	0.030	0.55 (0.27-1.13)	0.10
Peak mean PAP, mm Hg	1.74 (1.03-2.93)	0.039	Not selected	—	Not selected	—
Log peak PA compliance, mL/mm Hg	0.41 (0.23-0.74)	0.003	Not selected	—	Not selected	—
PCWP/CO slope	2.68 (0.996-7.18)	0.051	Not selected	—	Not selected	—
Mean PAP/CO slope	2.56 (1.30-5.04)	0.007	Not selected	—	Not selected	—
Rest stroke volume, mL	0.54 (0.33-0.90)	0.018	Not selected	—	Not selected	—
Peak stroke volume, mL	0.39 (0.20-0.76)	0.005	Not selected	—	Not selected	—
Peak workload, 10 W	0.80 (0.67-0.95)	0.012	Not selected	—	Not selected	—
Peak V _{O2} , mL/min/kg	0.61 (0.37-0.99)	0.048	Not selected	—	Not selected	—

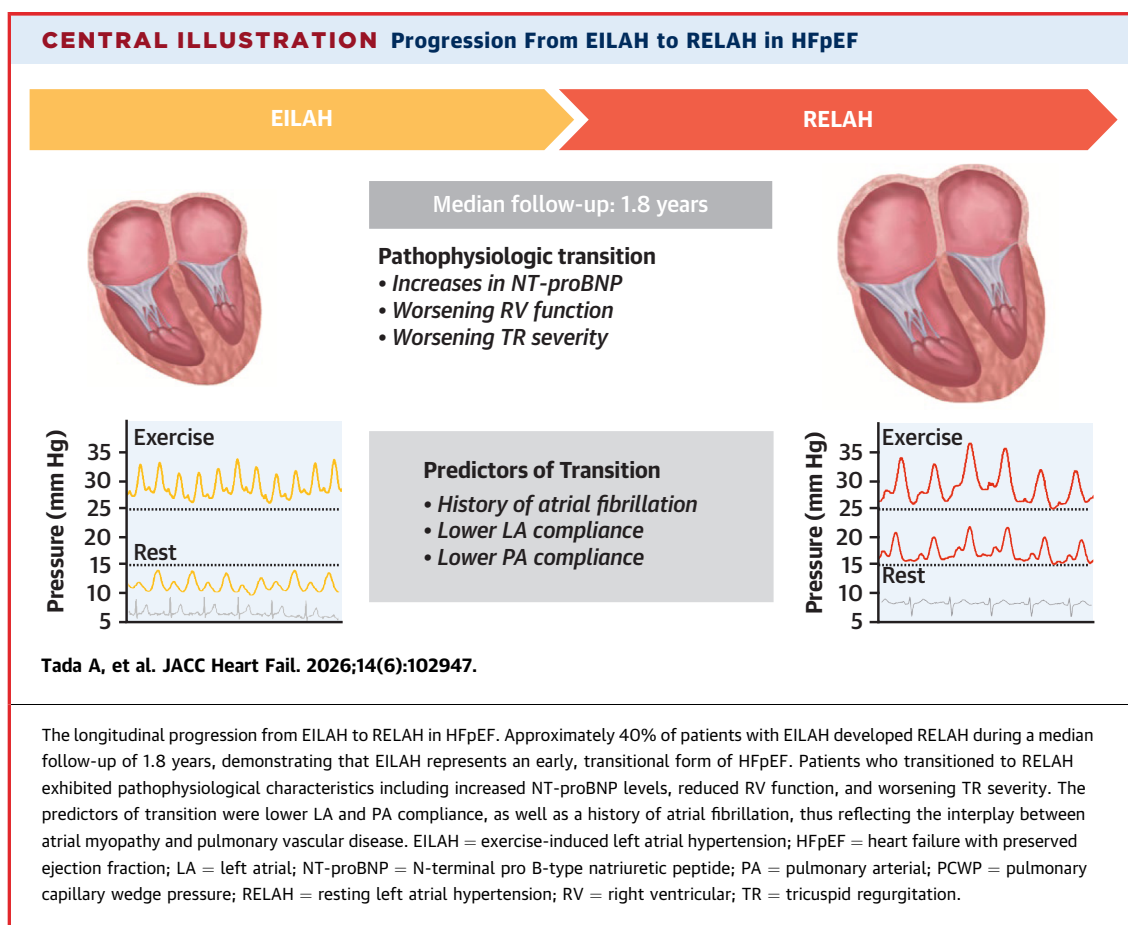
Maximum of 3 covariates included in multivariable models given the presence of 33 events.
Abbreviations as in [Tables 1 and 2](#).

(Dapagliflozin Effect on Cardiovascular Events-Thrombolysis In Myocardial Infarction 58) trial showed that dapagliflozin reduced the new onset of atrial fibrillation in patients with and without HF (another biomarker of atrial myopathy).²⁴ These studies suggest that SGLT2 inhibitors may help prevent transition from EILAH to RELAH through favorable effects on atrial, systemic, and pulmonary vascular compliance, thus pointing to potential roles for other cardiometabolic interventions in this setting.

Atrial fibrillation was an independent predictor of transition to RELAH, and this begs the question whether its treatment could mitigate this transition. There is limited evidence on the effectiveness of ablation for atrial fibrillation in patients with HFpEF,^{25,26} but 1 small mechanistic trial in HFpEF with atrial fibrillation showed improvement in exercise hemodynamics and aerobic capacity following catheter ablation, to the point where some of the participants no longer fulfilled the diagnostic criteria for EILAH following catheter ablation.²⁷ An ongoing randomized trial (CABA-HFPEF [Catheter-Based

Ablation of Atrial Fibrillation Compared to Conventional Treatment in Patients With Heart Failure With Preserved Ejection Fraction]; [NCT05508256](#)) is evaluating the effect of catheter ablation on hemodynamic and clinical outcomes in HFpEF with atrial fibrillation.²⁸ Conversely, atrial fibrosis from catheter ablation could worsen LA compliance. Further study in randomized trials is indicated.

STUDY LIMITATIONS. The cross-sectional nature of the data precludes drawing inferences toward causality in the associations between atriopulmonary compliance and transition to RELAH. The sample size is modest due to the requirement for multiple invasive studies, thus reducing power, but such data are singular in the published reports exploring longitudinal changes over time in invasive exercise hemodynamics. Since patient enrollment began in 2006, the use of contemporary HFpEF therapies such as SGLT2 inhibitors and glucagon-like peptide 1 receptor agonists has been very limited. Wider adoption of these medications in current practice may alter disease trajectories and hemodynamic responses observed in this study. The most significant limitation relates to



the observational design and potential selection bias because analyses were restricted to patients who underwent repeat invasive cardiopulmonary exercise testing. Only 13% of patients with EILAH underwent subsequent testing, and this subgroup represented a higher-risk group with poorer kidney function, greater diuretic agent use, and higher prevalence of diabetes and atrial fibrillation, and the observed transition rate should be interpreted within the context of the greater comorbidity burden of those patients with repeat evaluation. Follow-up duration was not uniform and was longer among patients who progressed to RELAH. However, multivariable logistic regression analysis examining the predictors of patients with change from EILAH to RELAH adjusted the observation duration as a covariate, and the finding that follow-up duration was greater in those patients developing RELAH is consistent with the fact that cardiac mechanics and arterial compliance worsen over time, so a greater duration of time thus allowed for greater exposure to

cause RELAH. In addition, although we intentionally avoided including closely related variables within the same multivariable models, residual collinearity among physiologically related measures cannot be fully excluded and may influence the estimated magnitude of individual associations. Some patients who did not transition to RELAH no longer met exercise hemodynamic criteria for HFpEF at follow-up, thus potentially reflecting lifestyle modification, optimization of volume status, or initiation of guideline-directed medical therapies such as SGLT2 inhibitors and mineralocorticoid receptor antagonists.²² Our study focused on forward transition from EILAH to RELAH, and future bidirectional analyses, including patients with RELAH at baseline, may better characterize the dynamic hemodynamic continuum in HFpEF. Finally, we did not attempt to determine specific cutoff values for LA or PA compliance to avoid potential overfitting in this relatively small cohort. Future larger studies are warranted to validate

clinically meaningful thresholds for these parameters and to determine their predictive utility in broader HFpEF groups, but this first report supports the concept hypothesis that impairments in LA and PA compliance are associated with progression to RELAH.

CONCLUSIONS

Significant numbers of patients with HFpEF who first present with EILAH transition to RELAH over time, a finding suggesting that EILAH may represent an earlier form of HFpEF. A history of atrial fibrillation and reductions in LA and PA compliance are independently associated with progression to RELAH, thereby suggesting a potential role for interventions that enhance atriopulmonary compliance as a promising treatment to mitigate disease progression in HFpEF.

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GlaxoSmithKline, Medtronic, Mesoblast, Novo Nordisk, Rivus, and Tenax Therapeutics; has served as a consultant for Actelion, Amgen, Aria, Axon Therapies, Becton Dickinson, Boehringer Ingelheim, Cytokinetics, Edwards Lifesciences, Eli Lilly, Imbria, Janssen, Merck Sharp and Dohme, Novo Nordisk, NGM, NXT, and VADovations; and has been named inventor (U.S. Patent No. 10,307,179) for the tools and approach for a minimally invasive pericardial modification procedure to treat heart failure. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: Progression from EILAH to RELAH represents a key transition in the natural history of HFpEF. Impaired LA and pulmonary vascular compliance identifies patients at higher risk, underscoring the pathophysiological continuum linking atrial myopathy and PVD in HFpEF.

TRANSLATIONAL OUTLOOK: Future studies should evaluate interventions that enhance atrial and pulmonary vascular compliance to mitigate or even reverse disease progression in HFpEF.

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KEY WORDS EILAH, heart failure, HFpEF, PCWP, RELAH

APPENDIX For supplemental tables, please see the online version of this paper.